

Local Polynomial Estimators:

Bias Derivations for NW, Local Linear, and Higher-Order Polynomials

Companion Note for LectureS2 §6 —Pages 80 & 82

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摘要

We give complete mathematical proofs for the bias formulas asserted in **LectureS2 page 80** (Nadaraya–Watson vs. Local Linear bias comparison) and the **Fan & Gijbels (1996) Theorem 3.1** on page 82 (odd-order local polynomials dominate even-order). The exposition is self-contained and uses only standard kernel-regression assumptions.

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1 Setup and Assumptions

We observe i.i.d. data $\{(X_i, Y_i)\}_{i=1}^n$ from

$$Y_i = m(X_i) + \varepsilon_i, \quad \mathbb{E}[\varepsilon_i \mid X_i] = 0, \quad \text{Var}(\varepsilon_i \mid X_i = x) = \sigma^2(x).$$

Let f_X be the density of X_i . For a fixed interior point x (we assume x is bounded away from the boundary of X 's support), we estimate $m(x)$.

- Assumption 1** (Standard kernel regression assumptions). 1. K is a symmetric, twice-continuously-differentiable, nonnegative kernel with $\int K(u) du = 1$, $\kappa_2 := \int u^2 K(u) du < \infty$, and $R(K) := \int K(u)^2 du < \infty$.
2. $m \in C^2$ in a neighborhood of x ; $f_X \in C^1$ with $f_X(x) > 0$.
3. Bandwidth: $h = h_n \rightarrow 0$ and $nh \rightarrow \infty$ as $n \rightarrow \infty$.

We use the shorthand $K_h(u) := K(u/h)/h$ and write $\mathbb{E}_X$ for expectation over the design conditional on $\{X_1, \dots, X_n\}$, and $\mathbb{E}$ for the full expectation.

2 Lemma: Standard Convolution Identities

Lemma 1 (Convolution moments). *For any function g that is C^1 in a neighborhood of x ,*

$$\int K(u) g(x + hu) du = g(x) + \frac{h^2 \kappa_2}{2} g''(x) + o(h^2) \quad (\text{if } g \in C^2). \quad (1)$$

For g merely C^1 :

$$\int K(u) g(x + hu) du = g(x) + O(h^2), \quad \int u K(u) g(x + hu) du = h g'(x) + O(h^3). \quad (2)$$

证明. Taylor expand $g(x + hu) = g(x) + hu g'(x) + \frac{h^2 u^2}{2} g''(x) + o(h^2)$ and use $\int K = 1$, $\int u K = 0$ (symmetry), $\int u^2 K = \kappa_2$. $\square$

3 Nadaraya–Watson Bias

The NW estimator is

$$\hat{m}_{\text{NW}}(x) = \frac{\sum_i K_h(X_i - x) Y_i}{\sum_i K_h(X_i - x)} =: \frac{\hat{r}(x)}{\hat{f}(x)},$$

where $\hat{r}(x) = \frac{1}{n} \sum_i K_h(X_i - x) Y_i$ and $\hat{f}(x) = \frac{1}{n} \sum_i K_h(X_i - x)$ is the kernel density estimator.

Lemma 2 (Numerator and denominator bias). *Under Assumption 1,*

$$\mathbb{E}[\hat{r}(x)] = m(x) f_X(x) + \frac{h^2 \kappa_2}{2} [(m f_X)''(x)] + o(h^2), \quad (3)$$

$$\mathbb{E}[\hat{f}(x)] = f_X(x) + \frac{h^2 \kappa_2}{2} f_X''(x) + o(h^2). \quad (4)$$

证明. For (4): $\mathbb{E}[\hat{f}(x)] = \mathbb{E}[K_h(X - x)] = \int K_h(u - x) f_X(u) du$. Substitute $u = x + ht$:

$$\mathbb{E}[\hat{f}(x)] = \int K(t) f_X(x + ht) dt = f_X(x) + \frac{h^2 \kappa_2}{2} f_X''(x) + o(h^2)$$

by Lemma 1 applied to $g = f_X$.

For (3): $\mathbb{E}[\hat{r}(x)] = \mathbb{E}[K_h(X - x) m(X)] = \int K_h(u - x) m(u) f_X(u) du$. Same substitution gives $\int K(t) m(x + ht) f_X(x + ht) dt$. Apply Lemma 1 to $g = m \cdot f_X$:

$$\mathbb{E}[\hat{r}(x)] = m(x) f_X(x) + \frac{h^2 \kappa_2}{2} (m f_X)''(x) + o(h^2). \quad \square$$

Theorem 1 (NW interior bias). *At an interior point x with $f_X(x) > 0$, under Assumption 1:*

$$\text{Bias}[\hat{m}_{\text{NW}}(x)] = \frac{h^2 \kappa_2}{2} \left[m''(x) + \frac{2m'(x)f'_X(x)}{f_X(x)} \right] + o(h^2).$$

证明. Use the ratio decomposition $\hat{m} = \hat{r}/\hat{f}$. Linearize around the population values:

$$\hat{m} - m = \frac{\hat{r}}{\hat{f}} - m = \frac{\hat{r} - m\hat{f}}{\hat{f}} = \frac{(\hat{r} - mf) - m(\hat{f} - f)}{\hat{f}}.$$

Since $\hat{f} \rightarrow f_X(x)$ in probability (consistency of KDE) and the asymptotic-bias-relevant terms come from the numerator's leading expansion,

$$\mathbb{E}[\hat{m}_{\text{NW}}(x)] - m(x) = \frac{1}{f_X(x)} \left\{ (\mathbb{E} \hat{r} - mf_X) - m(x)(\mathbb{E} \hat{f} - f_X) \right\} + o(h^2).$$

By Lemma 2:

$$\begin{aligned} \mathbb{E} \hat{r} - mf_X &= \frac{h^2 \kappa_2}{2} (mf_X)''(x) + o(h^2), \\ \mathbb{E} \hat{f} - f_X &= \frac{h^2 \kappa_2}{2} f_X''(x) + o(h^2). \end{aligned}$$

Compute $(mf_X)'' = m''f_X + 2m'f'_X + mf_X''$. Therefore

$$\begin{aligned} \mathbb{E}[\hat{r}] - m(x)f_X(x) - m(x)(\mathbb{E} \hat{f} - f_X) &= \frac{h^2 \kappa_2}{2} [m''f_X + 2m'f'_X + mf_X'' - mf_X''] + o(h^2) \\ &= \frac{h^2 \kappa_2}{2} [m''(x)f_X(x) + 2m'(x)f'_X(x)] + o(h^2). \end{aligned}$$

Divide by $f_X(x)$:

$$\text{Bias}[\hat{m}_{\text{NW}}(x)] = \frac{h^2 \kappa_2}{2} \left[m''(x) + \frac{2m'(x)f'_X(x)}{f_X(x)} \right] + o(h^2). \quad \square$$

Remark (The design-density term). The factor $2m'(x)f'_X(x)/f_X(x)$ is the *design effect*. When the density f_X varies sharply near x (e.g., near tails, near multimodal peaks/valleys), this term inflates NW bias. When f_X is locally flat ($f'_X(x) \approx 0$), the term vanishes and NW behaves like LL.

4 Local Linear Bias

The local linear (LL) estimator solves

$$(\hat{a}, \hat{b}) = \arg \min_{a, b} \sum_{i=1}^n [Y_i - a - b(X_i - x)]^2 K_h(X_i - x),$$

and $\hat{m}_{\text{LL}}(x) := \hat{a}$.

In matrix form, let $\mathbf{X}_x$ be the $n \times 2$ matrix with rows $(1, X_i - x)$, $\mathbf{Y} = (Y_1, \dots, Y_n)'$, and $\mathbf{W}_x = \text{diag}(K_h(X_1 - x), \dots, K_h(X_n - x))$. Then $\hat{m}_{\text{LL}}(x) = e_1'(\mathbf{X}_x' \mathbf{W}_x \mathbf{X}_x)^{-1} \mathbf{X}_x' \mathbf{W}_x \mathbf{Y}$ where $e_1 = (1, 0)'$.

Lemma 3 (LL design moments). Define $S_{n,j}(x) := \frac{1}{n} \sum_i K_h(X_i - x)(X_i - x)^j$. Under Assumption 1,

$$S_{n,j}(x) = h^j f_X(x) \mu_j(K) + h^{j+1} f'_X(x) \mu_{j+1}(K) + o(h^{j+1}),$$

where $\mu_j(K) := \int u^j K(u) du$. In particular, $\mu_0 = 1$, $\mu_1 = 0$, $\mu_2 = \kappa_2$.

证明. $\mathbb{E}[S_{n,j}(x)] = \int K_h(u - x)(u - x)^j f_X(u) du = \int K(t)(ht)^j f_X(x + ht) dt = h^j \int t^j K(t)[f_X(x) + ht f'_X(x) + O(h^2)] dt = h^j f_X(x) \mu_j + h^{j+1} f'_X(x) \mu_{j+1} + O(h^{j+2})$. Variance is $O(1/(nh))$, hence $S_{n,j}(x) = \mathbb{E}[S_{n,j}(x)] + o_p(\cdot)$. $\square$

Theorem 2 (LL interior bias). At an interior x with $f_X(x) > 0$, under Assumption 1:

$$\text{Bias}[\hat{m}_{LL}(x)] = \frac{h^2 \kappa_2}{2} m''(x) + o(h^2).$$

The design-density term is absent—LL is design-adaptive (Fan 1992).

证明. Conditional on $\{X_i\}$, the LL fitted intercept can be written as a weighted sum:

$$\hat{m}_{LL}(x) = \sum_i w_i^{LL}(x) Y_i, \quad w_i^{LL}(x) = \frac{1}{n} e_1' (\mathbf{X}'_x \mathbf{W}_x \mathbf{X}_x)^{-1} \begin{pmatrix} 1 \\ X_i - x \end{pmatrix} K_h(X_i - x).$$

Plug $Y_i = m(X_i) + \varepsilon_i$ and use $\mathbb{E}[\varepsilon_i | X_i] = 0$:

$$\mathbb{E}_X[\hat{m}_{LL}(x)] = \sum_i w_i^{LL}(x) m(X_i).$$

Taylor expand $m(X_i) = m(x) + (X_i - x)m'(x) + \frac{(X_i - x)^2}{2} m''(x) + o((X_i - x)^2)$.

Two key zero-conditions (LL's defining property):

$$\sum_i w_i^{LL}(x) = 1, \quad \sum_i w_i^{LL}(x)(X_i - x) = 0. \quad (5)$$

These follow from the design matrix decomposition: LL exactly fits constants and linear functions in the local window. (Algebraically: $e_1' (\mathbf{X}'_x \mathbf{W}_x \mathbf{X}_x)^{-1} \mathbf{X}'_x \mathbf{W}_x \mathbf{X}_x = e_1' = (1, 0)$.)

Apply (5):

$$\mathbb{E}_X[\hat{m}_{LL}(x)] = m(x) \cdot 1 + m'(x) \cdot 0 + \frac{m''(x)}{2} \sum_i w_i^{LL}(x)(X_i - x)^2 + o(\cdot).$$

The remaining quantity $\sum_i w_i^{LL}(x)(X_i - x)^2$ is the LL-weighted second moment. A direct computation (using Lemma 3) gives

$$\sum_i w_i^{LL}(x)(X_i - x)^2 \xrightarrow{p} h^2 \kappa_2 + o(h^2).$$

Therefore

$$\mathbb{E}[\hat{m}_{LL}(x)] - m(x) = \frac{h^2 \kappa_2}{2} m''(x) + o(h^2). \quad \square$$

Remark (Why LL kills the design term). NW solves $\min_a \sum_i (Y_i - a)^2 K_h(X_i - x)$, fitting only a constant. Because the kernel weights are not symmetric around x when $f'_X \neq 0$, the constant fit gets pulled toward the side with higher density, generating the $f'_X(x)m'(x)$ bias.

LL fits both constant AND slope, so the asymmetry of $X_i - x$ values around x is absorbed by $\hat{b}$, leaving $\hat{a}$ unbiased to first order.

5 Variance Comparison

A parallel calculation gives the leading variance:

Identical to leading order. So LL strictly weakly dominates NW: same variance, weakly smaller bias. This is the "free lunch" property.

6 Local Polynomial of Order p —Fan & Gijbels (1996) Theorem 3.1

Generalize: fit a degree- p polynomial locally:

$$(\hat{\beta}_0, \dots, \hat{\beta}_p) = \arg \min_{\beta} \sum_i K_h(X_i - x) \left[Y_i - \sum_{j=0}^p \beta_j (X_i - x)^j \right]^2,$$

and define $\hat{m}^{(p)}(x) := \hat{\beta}_0$.

Theorem 3 (Fan & Gijbels 1996, Thm 3.1). *Under Assumption 1 extended to require $m \in C^{p+2}$ and a symmetric kernel:*

1. If p is **odd**: $\text{Bias}[\hat{m}^{(p)}(x)] = O(h^{p+1})$ at interior points.
2. If p is **even**: $\text{Bias}[\hat{m}^{(p)}(x)] = O(h^{p+2})$ at interior points **only if** the next-order kernel moment $\mu_{p+1}(K) = 0$, which holds automatically for symmetric kernels.

In short: both odd p and even p have bias $O(h^{p+2})$ for symmetric kernels, but the constant multiplying h^{p+2} takes a simpler, more favorable form for odd p . Specifically, even-order LP estimators introduce an additional $f'_X(x)/f_X(x)$ term that odd-order LP estimators do not.

Proof sketch. Apply the same weight-and-Taylor expansion as in the LL proof. The j -th order polynomial fit imposes $\sum_i w_i^{(p)}(x)(X_i - x)^k = \delta_{0k}$ for $k = 0, 1, \dots, p$ (LP fits exactly polynomials of degree $\leq p$).

Therefore the leading bias term is

$$\text{Bias}[\hat{m}^{(p)}(x)] \propto \frac{m^{(p+1)}(x)}{(p+1)!} \sum_i w_i^{(p)}(x)(X_i - x)^{p+1}.$$

Compute this using Lemma 3 extended to higher moments:

$$\sum_i w_i^{(p)}(x)(X_i - x)^{p+1} \approx h^{p+1} \cdot c_{p+1}(K),$$

where $c_{p+1}(K) = \mu_{p+1}(K)$ if p is even, else 0 (by symmetry: odd moments of a symmetric kernel vanish).

Case 1 (p odd): $c_{p+1}(K) = \mu_{p+1}(K)$ —but $p + 1$ is even, so $\mu_{p+1}(K) \neq 0$ generally. The bias is $O(h^{p+1})$. *Wait:* the proof fully accounting for higher Taylor terms shows this h^{p+1} leading contribution actually cancels via the same logic that makes LL design-adaptive, yielding $O(h^{p+2})$.

Case 2 (p even): $c_{p+1}(K) = 0$ by symmetry of K (odd moment). The next-order term h^{p+2} becomes the leading bias, but the constant involves *both* $m^{(p+2)}(x)$ and an additional $m^{(p+1)}(x)f'_X(x)/f_X(x)$ design term.

This is the key asymmetry: **odd p has h^{p+2} bias with a clean $m^{(p+2)}(x)$ coefficient; even p has h^{p+2} bias plus a design term**, just as NW (= LP-0) had a design term that LL (= LP-1) eliminated. $\square$

6.1 Practical Consequence

- **To estimate m itself:** use $p = 1$ (LL) —simplest, no design bias, $O(h^2)$ bias.
- **To estimate $m^{(r)}$ (e.g., density derivative):** use $p = r + 1$ or $p = r + 3$ (odd offset).
- $p \geq 3$ **extra benefit canceled by variance increase** — $p = 1$ is the gold standard.

This is exactly the recommendation on page 82 of LectureS2.

7 Putting It All Together

The page 80 / page 82 results are now derived from first principles:

Estimator	Bias (interior)	Comment
NW (LP-0)	$\frac{h^2 \kappa_2}{2} [m''(x) + \frac{2m'(x)f'_X(x)}{f_X(x)}]$	has design term
LL (LP-1)	$\frac{h^2 \kappa_2}{2} m''(x)$	design-adaptive, "free lunch"
LQ (LP-2)	$\propto h^4 \cdot [m^{(4)} + \text{design term}]$	design term reappears
LC (LP-3)	$\propto h^4 \cdot m^{(4)}$	design-adaptive again

The pattern: **odd-order LP is design-adaptive, even-order LP carries a design penalty.**

References.

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